DAY 3 — HOUSE PRICE DATA VISUALISATION

Graphs + interpretation | Based on the uploaded train_HousingPrice.csv

Dataset

1,460 houses
81 columns

Target

SalePrice
House sale price

Focus

Choose the chart
Interpret the pattern

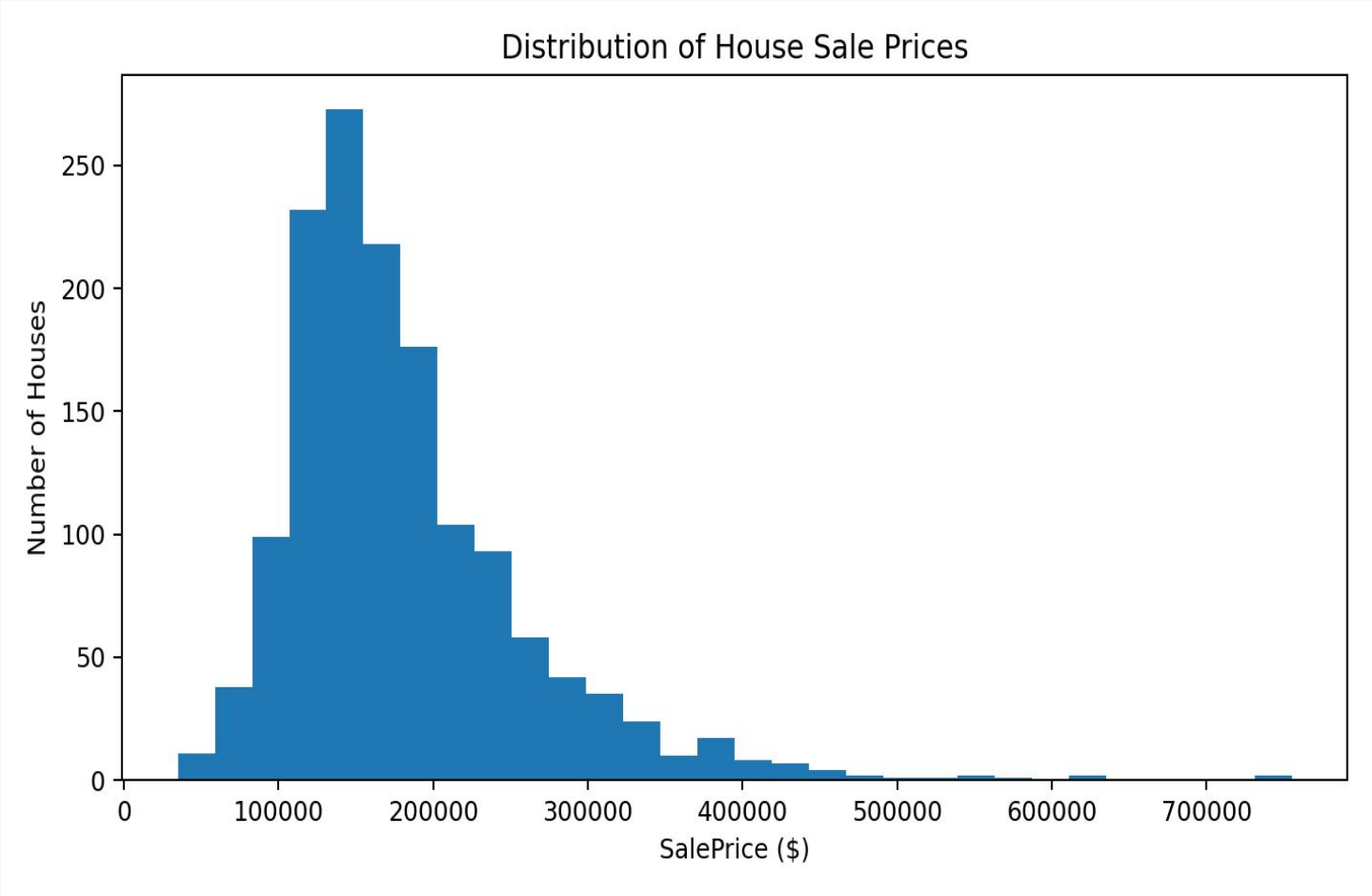
Visualisation set

Histogram • Column • Scatter • Bar • Line • Boxplot • Correlation heatmap

Key rule

The business question determines the visualisation.

1. Histogram — SalePrice



Interpretation

The distribution is strongly right-skewed. Many houses are concentrated in lower-to-middle price ranges, while fewer expensive houses create a long right tail. High-value observations can pull the mean upward, so median and distribution summaries are useful.

Classroom question: Is the distribution symmetric? What does the right tail indicate?

2. Column Chart — OverallQual vs Average SalePrice



Interpretation

Average SalePrice generally increases as OverallQual increases. This indicates a strong association between quality rating and price in this dataset. It does not prove that quality alone causes the price difference.

Classroom question: Do higher quality ratings correspond to higher average prices?

3. Scatter Plot — GrLivArea vs SalePrice

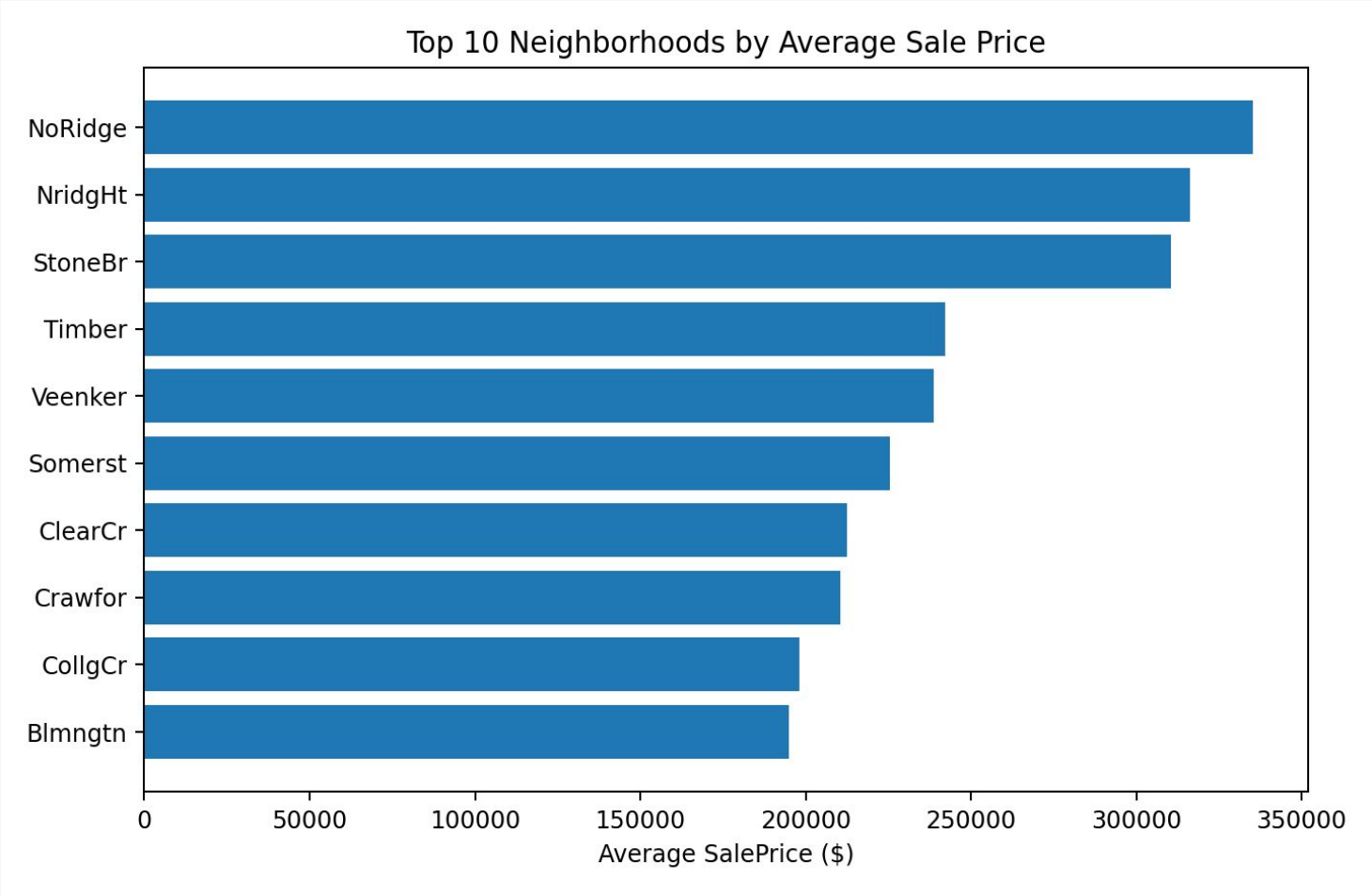


Interpretation

There is a clear positive relationship: larger above-ground living areas tend to be associated with higher SalePrice. The points are not perfectly aligned, showing that other property characteristics also matter. A few observations lie far from the main pattern.

Classroom question: What happens to price as living area increases? Are there potential outliers?

4. Bar Chart — Neighborhood vs Average Price

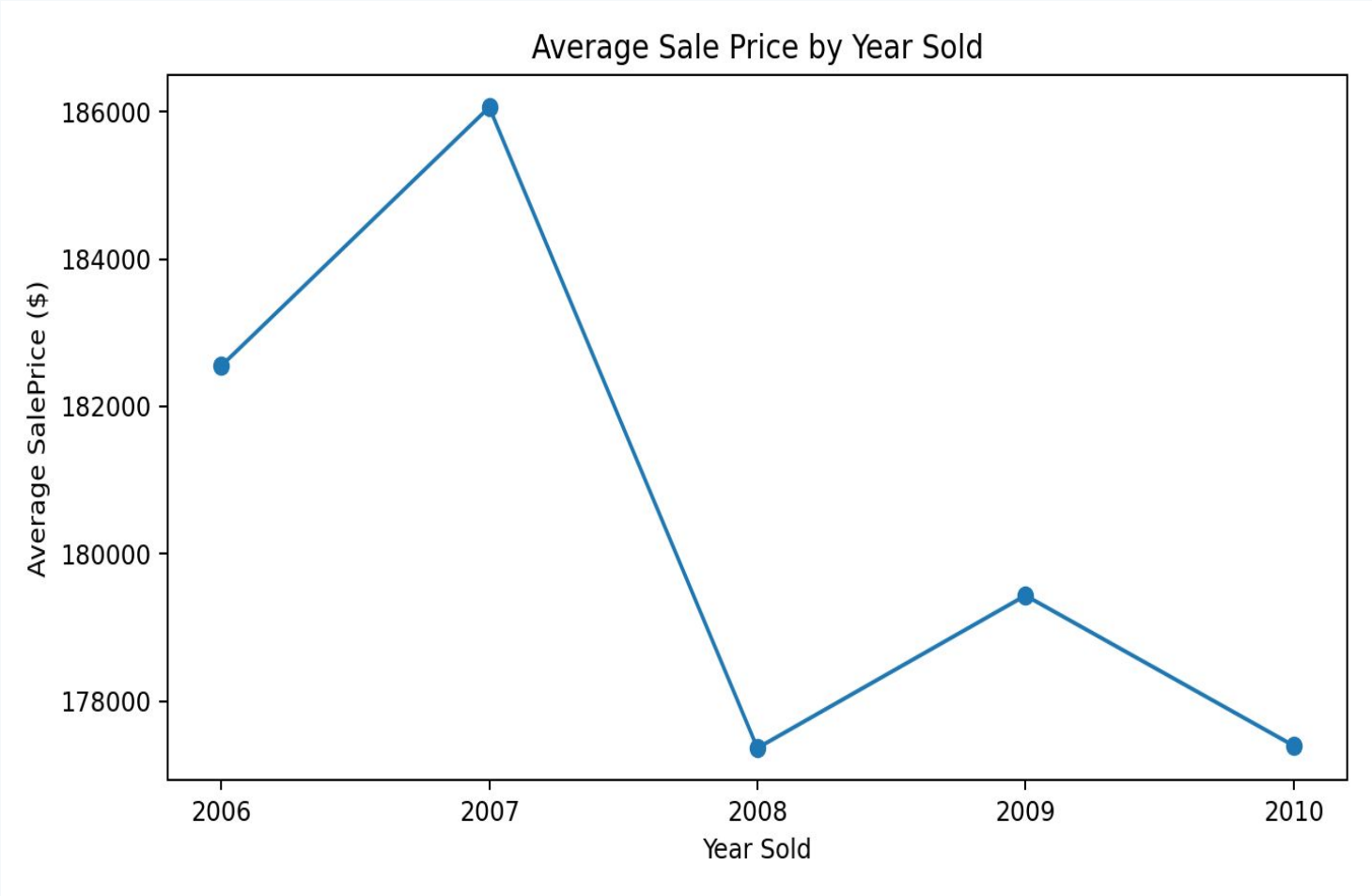


Interpretation

Average sale prices vary substantially among neighborhoods. Location is therefore important for exploratory analysis. However, average price should be considered together with the number of houses represented by each neighborhood.

Classroom question: Which neighborhoods have higher averages? Why should sample size also be checked?

5. Line Chart — Average SalePrice by Year

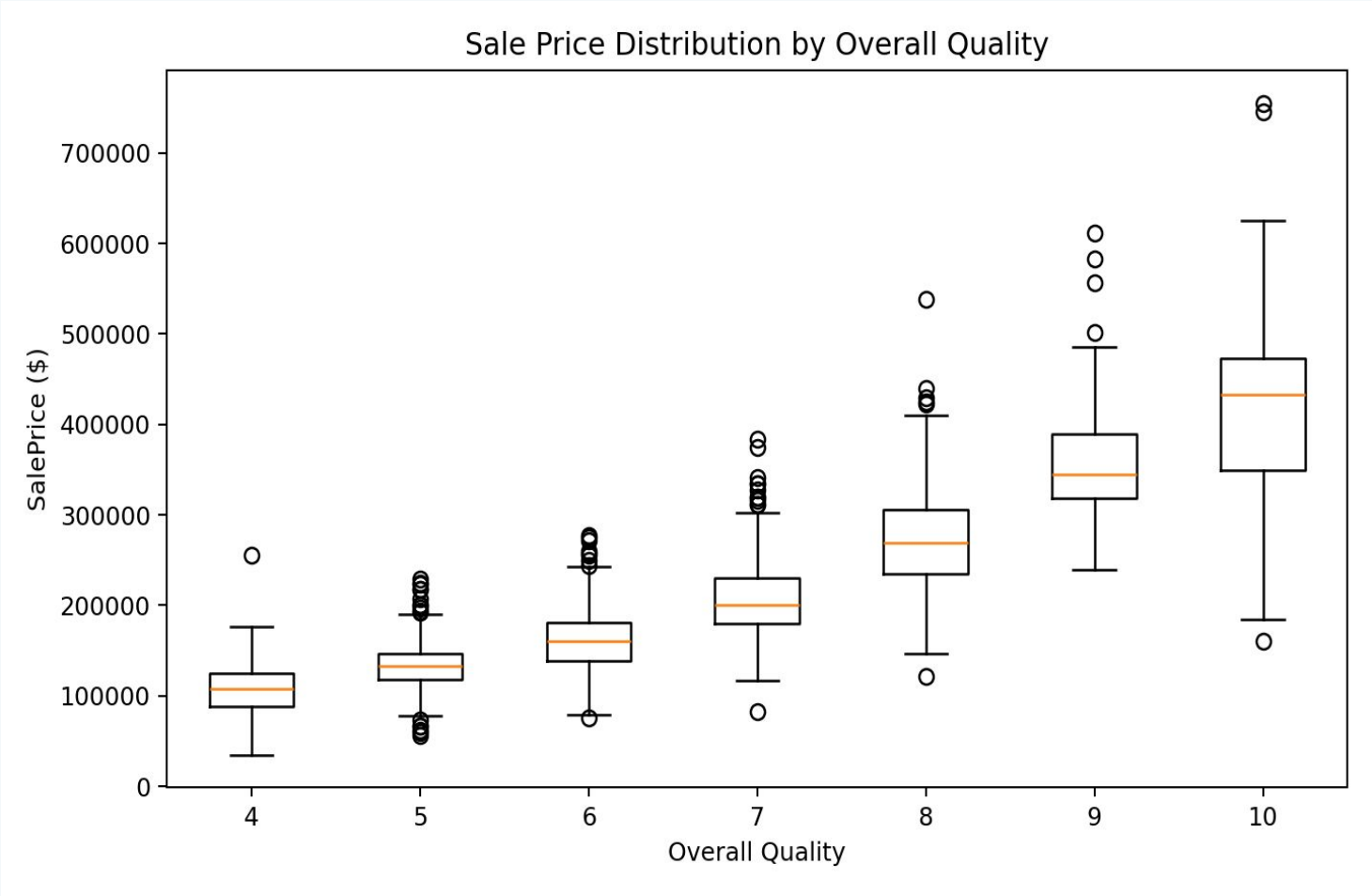


Interpretation

Average SalePrice changes across the sale years, with both increases and decreases rather than a simple continuously rising pattern. The line chart makes the sequence over time easy to inspect.

Classroom question: Which years show increases or decreases? Is there a consistent trend?

6. Boxplot — SalePrice by Overall Quality

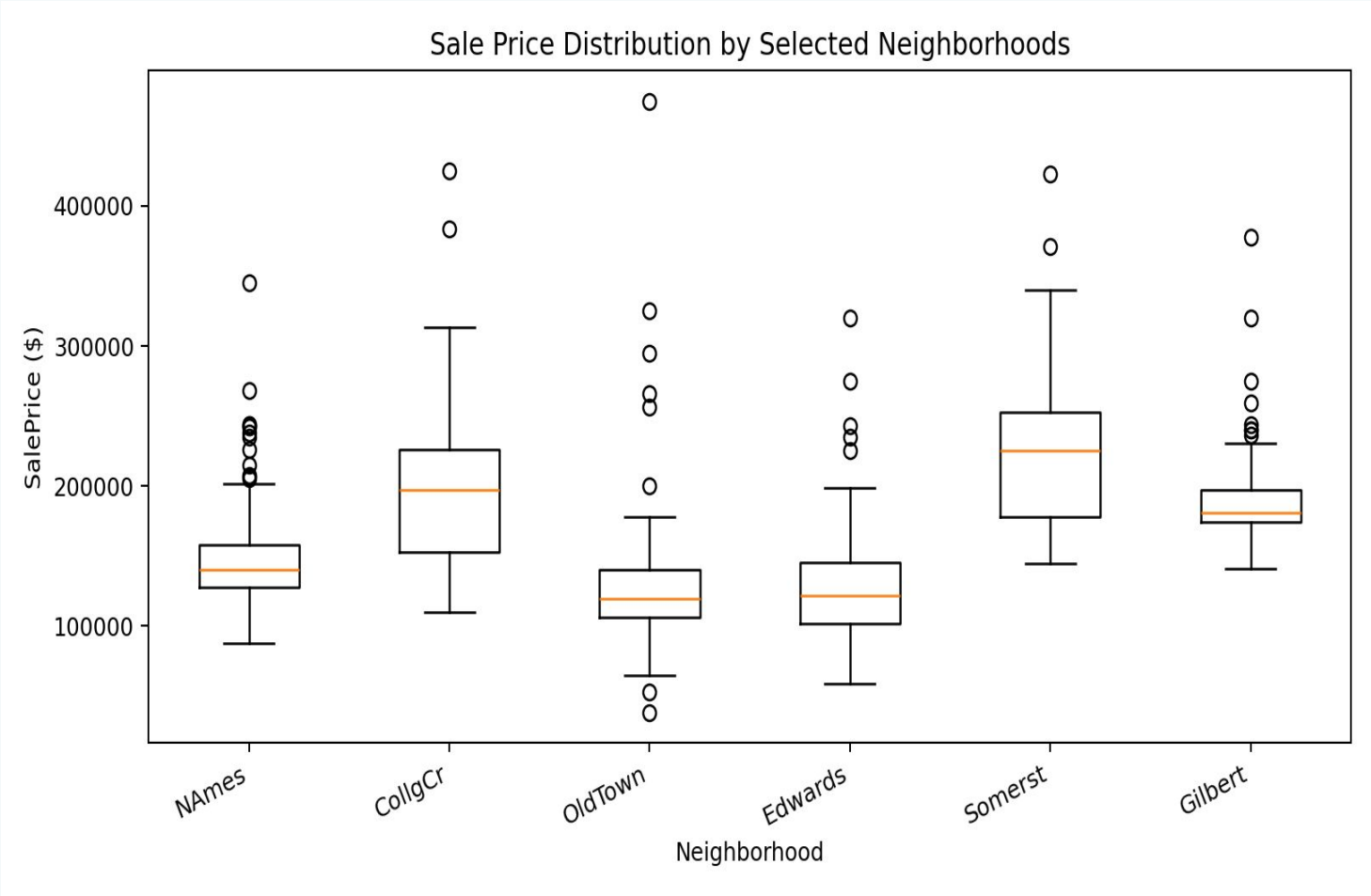


Interpretation

The median SalePrice generally shifts upward with OverallQual. The boxes show the middle 50% of prices, while whiskers and individual points show spread and potential outliers. The boxplot reveals distributional differences that an average-only chart can hide.

Classroom question: Which group has the highest median? Which has the greatest spread or outliers?

7. Boxplot — Selected Neighborhoods

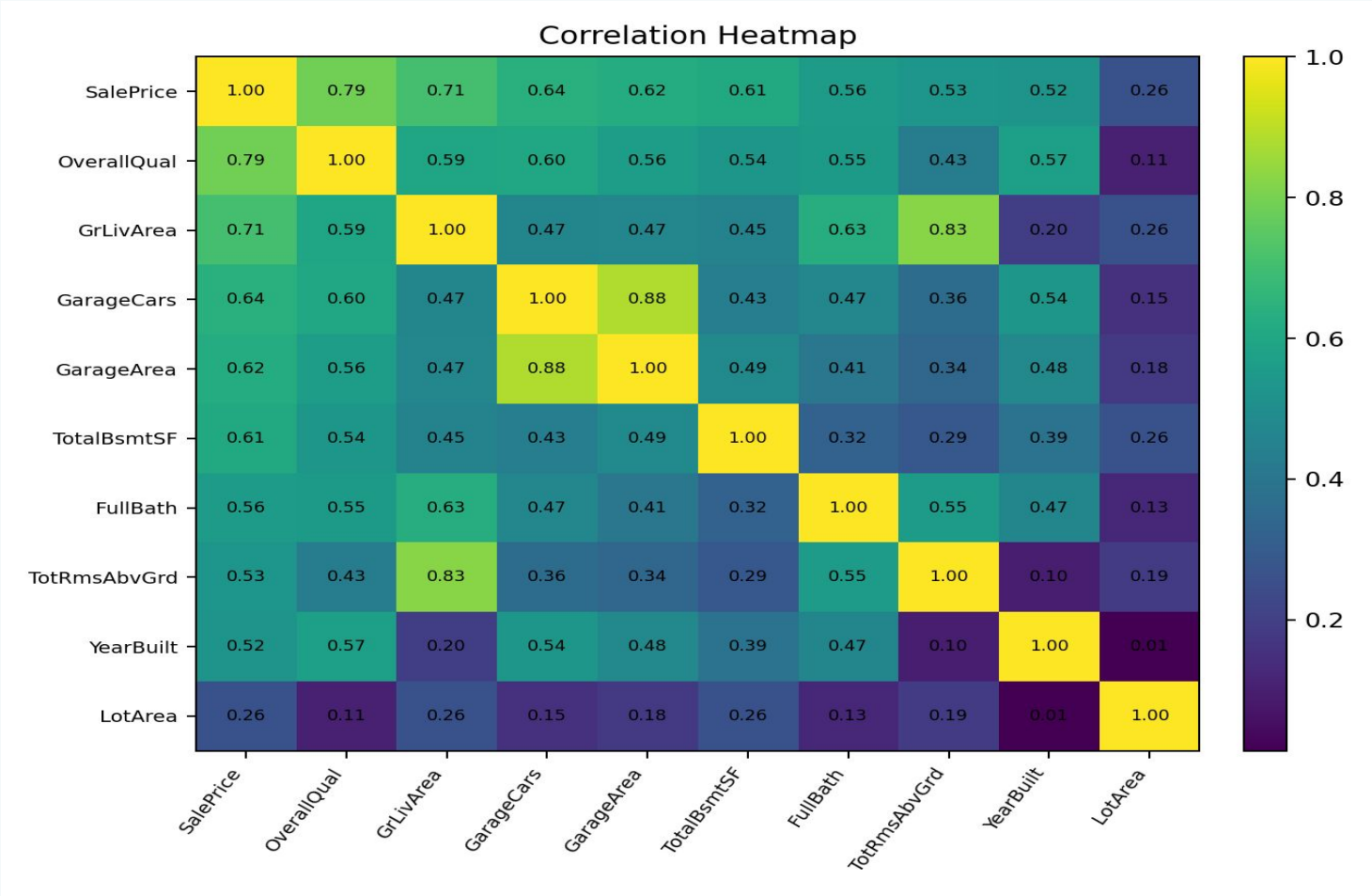


Interpretation

Neighborhoods differ in median price and variability. A wider box indicates a larger interquartile range. Points beyond the whiskers are potential outliers and should be investigated rather than automatically treated as errors.

Classroom question: Which neighborhood has the highest median? Which has the widest distribution?

8. Correlation Heatmap — Numerical Variables



Interpretation

OverallQual shows a strong positive relationship with SalePrice. GrLivArea, GarageCars, GarageArea and TotalBsmtSF also show positive relationships. Correlation measures linear association and does not establish causation.

Classroom question: Which variables appear most strongly associated with SalePrice?

How to Read a Boxplot

Five important ideas

Median — line inside the box
Q1 — 25th percentile
Q3 — 75th percentile
IQR = Q3 − Q1
Whiskers — typical range
Potential outliers — points beyond whiskers

Why use it?

Compare groups using median, variability and outliers — not just average price.

Excel

Arrange SalePrice values by group → Insert → Statistical Chart → Box & Whisker.

How Students Should Interpret a Graph

1. Describe

What do you see?

Example:
“Average price increases as OverallQual increases.”

2. Explain carefully

What might the pattern suggest?

Avoid claiming causation from a chart alone.

3. Investigate

What should we check next?

Sample size • outliers • correlation • other variables

Good interpretation

Pattern + Evidence + Caution

Day 3 Classroom Activity

Task 1

Create:

- SalePrice histogram
- OverallQual column chart
- GrLivArea scatter plot

Task 2

Create:

- Neighborhood bar chart
- YrSold line chart
- OverallQual boxplot

Task 3

For each chart write:

1. Pattern
2. Interpretation
3. One limitation or question for further analysis

Student deliverable

6 charts + 6 interpretations + one observation about an outlier.

Key Takeaways

Chart → Question

Histogram → distribution
Bar / Column → category comparison
Line → time trend
Scatter → numerical relationship
Boxplot → median + spread + outliers
Heatmap → correlations

Visualisation → Analysis

Visualisation helps discover patterns.

Next:
→ Descriptive statistics
→ Correlation
→ Outlier investigation
→ Feature selection
→ Regression / ML

Final classroom question

What does the boxplot reveal that the average price alone cannot show?

1. Histogram — SalePrice — Excel Formula & Plotting Steps

Purpose	Understand the distribution of house prices and identify skewness and extreme values.
Excel formula	=COUNTIFS($H$2:$H$1461,">="&A2,$H$2:$H$1461,"<"&A3)
Alternative summary formulas	=AVERAGE(H2:H1461) | =MEDIAN(H2:H1461) | =MIN(H2:H1461) | =MAX(H2:H1461)
Steps to plot	1. Select SalePrice column. 2. Insert → Statistic Chart → Histogram. 3. Set Bin Width or Number of Bins. 4. Add chart title and axis titles. 5. Format the x-axis as currency if required.
Interpretation	A long right tail indicates that a smaller number of expensive houses pull the distribution upward. Compare mean and median when discussing skewness.

2. Column Chart — OverallQual vs Average SalePrice — Excel Formula & Plotting Steps

Purpose

Compare average selling price across house-quality ratings.

Excel formula

=AVERAGEIF($D$2:$D$1461,A2,$H$2:$H$1461)

Useful count

=COUNTIF($D$2:$D$1461,A2)

Steps to plot

1. Create a small summary table with OverallQual 1–10.
2. Calculate Average SalePrice for each rating.
3. Select the two summary columns.
4. Insert → Column Chart → Clustered Column.
5. Add data labels and a clear title.

Interpretation

Average SalePrice generally rises as OverallQual increases. This is an association; it does not by itself prove that quality causes the price difference.

3. Scatter Plot — GrLivArea vs SalePrice — Excel Formula & Plotting Steps

Purpose	Examine the relationship between above-ground living area and sale price.
Excel formula	No aggregation is required. Use the original GrLivArea and SalePrice observations directly.
Optional correlation	=CORREL($F$2:$F$1461,$H$2:$H$1461)
Steps to plot	1. Select GrLivArea and SalePrice. 2. Insert → Scatter → Scatter with only markers. 3. Put GrLivArea on the X-axis and SalePrice on the Y-axis. 4. Add a linear trendline. 5. Display R^2 if desired.
Interpretation	

The plot shows a positive relationship: larger living areas tend to be associated with higher prices.
Scatter around the trend shows that other factors also influence price.

4. Bar Chart — Neighborhood vs Average SalePrice — Excel Formula & Plotting Steps

Purpose Compare average house prices across neighborhoods.

Excel formula =AVERAGEIF($E$2:$E$1461,A2,$H$2:$H$1461)

Count by neighborhood =COUNTIF($E$2:$E$1461,A2)

Steps to plot

1. Create a neighborhood summary table.
2. Calculate Average SalePrice and Count.
3. Sort Average SalePrice from largest to smallest.
4. Select the top 10 neighborhoods.
5. Insert → Bar Chart → Clustered Bar.

Interpretation

Average prices differ substantially by neighborhood.
Always check the number of observations because an average based on very few houses can be less stable.

5. Line Chart — Average SalePrice by Year Sold — Excel Formula & Plotting Steps

Purpose

Examine how average sale price changes across sale years.

Excel formula

=AVERAGEIF($C$2:$C$1461,A2,$H$2:$H$1461)

Count by year

=COUNTIF($C$2:$C$1461,A2)

Steps to plot

1. Extract or list YrSold values.
2. Calculate average SalePrice for each year.
3. Sort years chronologically.
4. Select Year and Average SalePrice.
5. Insert → Line Chart.
6. Add markers and axis titles.

Interpretation

The line chart shows year-to-year changes in average SalePrice.
Avoid interpreting a short time series as a long-term trend without additional evidence.

6. Boxplot — SalePrice by OverallQual — Excel Formula & Plotting Steps

Purpose	Compare the median, spread and potential outliers of SalePrice across quality ratings.
Excel preparation	For each OverallQual value, filter/copy the corresponding SalePrice values into separate columns, or use FILTER in newer Excel versions.
Dynamic FILTER example	=FILTER($H$2:$H$1461,$D$2:$D$1461=A1)
Steps to plot	1. Create one SalePrice column for each selected OverallQual. 2. Select the grouped columns. 3. Insert → Statistic Chart → Box & Whisker. 4. Add title and axis labels.
Interpretation	5. Compare medians, IQRs and outliers.

Higher quality groups generally have higher median prices. The boxplot also reveals within-group variability that an average alone cannot show.

7. Boxplot — SalePrice by Neighborhood — Excel Formula & Plotting Steps

Purpose

Compare the full price distributions across selected neighborhoods.

Excel preparation

Create separate SalePrice columns for selected neighborhoods using FILTER or by filtering the source data.

Dynamic FILTER example

=FILTER($H$2:$H$1461,$E$2:$E$1461=A1)

Steps to plot

1. Choose a manageable number of neighborhoods.
2. Put each neighborhood's SalePrice values in a separate column.
3. Select the columns.
4. Insert → Statistic Chart → Box & Whisker.

Interpretation

5. Compare medians, IQRs, whiskers and potential outliers.

Neighborhoods can differ in both typical price and variability. Points beyond whiskers are potential outliers, not automatically data errors.

8. Correlation Heatmap — Numerical Variables — Excel Formula & Plotting Steps

Purpose

Explore linear relationships among important numerical variables.

Excel formula

=CORREL($D$2:$D$1461,$H$2:$H$1461)

Example pairs

OverallQual ↔ SalePrice
GrLivArea ↔ SalePrice
TotalBsmtSF ↔ SalePrice
GarageCars ↔ SalePrice

Steps to plot

1. Select numerical variables.
2. Build a correlation matrix using CORREL for each pair.
3. Select the matrix.
4. Home → Conditional Formatting → Color Scales.

Interpretation

5. Use the resulting matrix as a heatmap.

Positive values indicate positive linear association and negative values indicate negative linear association. Correlation does not establish causation.

Excel Formula Cheat Sheet for Day 3 Visualisation

Average by category	`=AVERAGEIF(criteria_range,criteria,average_range)`
Count by category	`=COUNTIF(criteria_range,criteria)`
Conditional average	`=AVERAGEIFS(avg_range,criteria_range1,criteria1,...)`
Correlation	`=CORREL(array1,array2)`
Percentile	`=PERCENTILE.INC(range,0.95)`
Quartile	`=QUARTILE.INC(range,1)` / `=QUARTILE.INC(range,3)`
Filter values	`=FILTER(array,include)`
Year from date	`=YEAR(date_cell)`

Day 3 Excel Workflow: Data → Formula → Chart → Interpretation

1. Identify	Choose the business question and variables.
2. Prepare	Check data types, blanks, duplicates and outliers.
3. Calculate	Use AVERAGEIF, COUNTIF, CORREL, FILTER and related formulas.
4. Summarize	Build a small analysis table before plotting.
5. Plot	Insert the chart type that matches the question.
6. Interpret	Describe the pattern, support it with evidence, and state a caution.